

ACR21455

## Dichloromethane-d2, for NMR, 99.8 atom % D

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 氘代二氯甲烷-D2  
 Dichloromethane-d2, for NMR, 99.8 atom % D

**Cat No. :** 214550000; 214550050; 214550100  
**CAS No** 1665-00-5  
**Molecular Formula** C Cl2 D2

**Supplier**
  
**UK entity/business name**  
 Fisher Scientific UK  
 Bishop Meadow Road,  
 Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
 Thermo Fisher Scientific  
 Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
 Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
sweet

#### Emergency Overview

Causes skin irritation. Causes serious eye irritation. May cause drowsiness and dizziness. Suspected of causing cancer. Hygroscopic.

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Carcinogenicity                                    | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements



# SAFETY DATA SHEET

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**Signal Word****Warning****Hazard Statements**

H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H336 - May cause drowsiness or dizziness  
H351 - Suspected of causing cancer

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection

**Response**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Hygroscopic.

**Health Hazards**

Causes skin irritation. Causes serious eye irritation. May cause drowsiness or dizziness. Suspected of causing cancer.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its volatility. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces.

Contains a known or suspected endocrine disruptor. Contains a substance on the National Authorities Endocrine Disruptor Lists.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component            | CAS No    | Weight % |
|----------------------|-----------|----------|
| Dichloro(2H2)methane | 1665-00-5 | 100      |
| Methylene chloride   | 75-09-2   | -        |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

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**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Use personal protective equipment as required.

**Notes to Physician**

A patient adversely affected by exposure to this product should not be given adrenaline (epinephrine) or similar heart stimulant since these would increase the risk of cardiac arrhythmias. Treat symptomatically. Symptoms may be delayed.

### SECTION 5. FIRE-FIGHTING MEASURES

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

**Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

### SECTION 7. HANDLING AND STORAGE

**Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere. Protect from moisture.

**Specific Use(s)**

Use in laboratories

### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

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### Control Parameters

| Component          | China                      | Taiwan                                    | Thailand                     | Hong Kong                                 |
|--------------------|----------------------------|-------------------------------------------|------------------------------|-------------------------------------------|
| Methylene chloride | TWA: 200 mg/m <sup>3</sup> | TWA: 50 ppm<br>TWA: 174 mg/m <sup>3</sup> | STEL: 125 ppm<br>TWA: 25 ppm | TWA: 50 ppm<br>TWA: 174 mg/m <sup>3</sup> |

| Component          | ACGIH TLV   | OSHA PEL                                                                                                          | NIOSH          | The United Kingdom                                                                                                         | European Union                                                                                                               |
|--------------------|-------------|-------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Methylene chloride | TWA: 50 ppm | (Vacated) TWA: 500 ppm<br>(Vacated) STEL: 2000 ppm<br>(Vacated) Ceiling: 1000 ppm<br>TWA: 25 ppm<br>STEL: 125 ppm | IDLH: 2300 ppm | STEL: 200 ppm 15 min<br>STEL: 706 mg/m <sup>3</sup> 15 min<br>TWA: 353 mg/m <sup>3</sup> 8 hr<br>TWA: 100 ppm 8 hr<br>Skin | TWA: 353 mg/m <sup>3</sup> (8h)<br>TWA: 100 ppm (8h)<br>STEL: 706 mg/m <sup>3</sup> (15min)<br>STEL: 200 ppm (15min)<br>Skin |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists  
 OSHA - Occupational Safety and Health Administration  
 NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

#### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371

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## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

## Environmental exposure controls

Do not allow material to contaminate ground water system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         |                                              |                                          |
|-----------------------------------------|----------------------------------------------|------------------------------------------|
| Appearance                              | Colorless                                    |                                          |
| Physical State                          | Liquid                                       |                                          |
| Odor                                    | sweet                                        |                                          |
| Odor Threshold                          | No data available                            |                                          |
| pH                                      | No information available                     |                                          |
| Melting Point/Range                     | -97 °C / -142.6 °F                           |                                          |
| Softening Point                         | No data available                            |                                          |
| Boiling Point/Range                     | 40 °C / 104 °F                               | @ 760 mmHg                               |
| Flash Point                             | No information available                     | <b>Method -</b> No information available |
| Evaporation Rate                        | No data available                            |                                          |
| Flammability (solid,gas)                | Not applicable                               | Liquid                                   |
| Explosion Limits                        | <b>Lower</b> 13 Vol%<br><b>Upper</b> 22 Vol% |                                          |
| Vapor Pressure                          | 450 hPa @ 20 °C                              |                                          |
| Vapor Density                           | No data available                            | (Air = 1.0)                              |
| Specific Gravity / Density              | 1.360                                        |                                          |
| Bulk Density                            | Not applicable                               | Liquid                                   |
| Water Solubility                        | Insoluble                                    |                                          |
| Solubility in other solvents            | No information available                     |                                          |
| Partition Coefficient (n-octanol/water) |                                              |                                          |
| Component                               | <b>log Pow</b>                               |                                          |
| Methylene chloride                      | 1.25                                         |                                          |
| Autoignition Temperature                | 556 °C / 1032.8 °F                           |                                          |
| Decomposition Temperature               | 120 °C                                       |                                          |
| Viscosity                               | No data available                            |                                          |
| Explosive Properties                    | No information available                     |                                          |
| Oxidizing Properties                    | No information available                     |                                          |
| Molecular Formula                       | C Cl2 D2                                     |                                          |
| Molecular Weight                        | 86.95                                        |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                                                     |
|--------------------------|---------------------------------------------------------------------|
| Stability                | Hygroscopic.                                                        |
| Hazardous Reactions      | None under normal processing.                                       |
| Hazardous Polymerization | Hazardous polymerization does not occur.                            |
| Conditions to Avoid      | Incompatible products. Excess heat. Exposure to moist air or water. |
| Materials to avoid       | Strong oxidizing agents. Strong acids. Amines.                      |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Phosgene. Hydrogen chloride gas.

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## SECTION 11. TOXICOLOGICAL INFORMATION

### Product Information

#### (a) acute toxicity;

| Component          | LD50 Oral            | LD50 Dermal          | LC50 Inhalation                                            |
|--------------------|----------------------|----------------------|------------------------------------------------------------|
| Methylene chloride | > 2000 mg/kg ( Rat ) | > 2000 mg/kg ( Rat ) | 53 mg/L ( Rat ) 6 h<br>76000 mg/m <sup>3</sup> ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

#### (d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component          | EU | UK | Germany | IARC     |
|--------------------|----|----|---------|----------|
| Methylene chloride |    |    |         | Group 2A |

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs

Central nervous system (CNS)

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs

None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

#### Other Adverse Effects

Tumorigenic effects have been reported in experimental animals.

#### Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

## SECTION 12. ECOLOGICAL INFORMATION

### Ecotoxicity effects

This product contains the following substance(s) which are hazardous for the environment.

| Component            | Freshwater Fish                           | Water Flea         | Freshwater Algae   | Microtox                                    |
|----------------------|-------------------------------------------|--------------------|--------------------|---------------------------------------------|
| Dichloro(2H2)methane | Pimephales promelas:<br>LC50:193 mg/L/96h | EC50: 140 mg/L/48h | EC50:>660 mg/L/96h | EC50: 1 mg/L/24 h<br>EC50: 2.88 mg/L/15 min |
| Methylene chloride   | Pimephales promelas:<br>LC50:193 mg/L/96h | EC50: 140 mg/L/48h | EC50:>660 mg/L/96h | EC50: 1 mg/L/24 h<br>EC50: 2.88 mg/L/15 min |

### Persistence and Degradability

Persistence

Persistence is unlikely, based on information available.

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**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component          | log Pow | Bioconcentration factor (BCF) |
|--------------------|---------|-------------------------------|
| Methylene chloride | 1.25    | 6.4 - 40 dimensionless        |

**Mobility in soil** The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

### SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with federal, state and local regulations. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Do not reuse empty containers. Dispose of in accordance with local regulations. Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

### SECTION 14. TRANSPORT INFORMATION

#### Road and Rail Transport

**UN-No** UN1593  
**Proper Shipping Name** DICHLOROMETHANE  
**Hazard Class** 6.1  
**Packing Group** III

#### IMDG/IMO

**UN-No** UN1593  
**Proper Shipping Name** DICHLOROMETHANE  
**Hazard Class** 6.1  
**Packing Group** III

#### IATA

**UN-No** UN1593  
**Proper Shipping Name** DICHLOROMETHANE  
**Hazard Class** 6.1  
**Packing Group** III

**Special Precautions for User** No special precautions required

### SECTION 15. REGULATORY INFORMATION

#### **International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The | List of | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|-----------|-----|---------|------|-------|--------|------|-----|-------|------|------|------|------|
|-----------|-----|---------|------|-------|--------|------|-----|-------|------|------|------|------|

# SAFETY DATA SHEET

## Dichloromethane-d2, for NMR, 99.8 atom % D

|                      |                                                 |                                 |   |   |           |   |   |   |   |   |   |          |
|----------------------|-------------------------------------------------|---------------------------------|---|---|-----------|---|---|---|---|---|---|----------|
|                      | Inventory of Hazardous Chemicals (2015 Edition) | dangerous goods GB 12268 - 2012 |   |   |           |   |   |   |   |   |   |          |
| Dichloro(2H2)methane | -                                               | -                               | X | - | 216-776-0 | - | - | - | - |   | - | -        |
| Methylene chloride   | X                                               | X                               | X | X | 200-838-9 | X | X | X | X | X | X | KE-23893 |

### National Regulations

| Component                         | Toxic Chemical Substances Control Act |
|-----------------------------------|---------------------------------------|
| Methylene chloride<br>75-09-2 (-) | Class IV (25 wt%)                     |

### SECTION 16. OTHER INFORMATION

**Creation Date** 19-Aug-2010  
**Revision Date** 07-Apr-2024  
**Revision Summary** Not applicable.

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
 Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
 First aid for chemical exposure, including the use of eye wash and safety showers.  
 Chemical incident response training.

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists  
**DNEL** - Derived No Effect Level  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**ENCS** - Japanese Existing and New Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances  
**NZIoC** - New Zealand Inventory of Chemicals

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
**PNEC** - Predicted No Effect Concentration  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
 Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS



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Dichloromethane-d<sub>2</sub>, for NMR, 99.8 atom % D

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## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**